Base station physics: summary

Params: base/spec/params.py v1.3.1

Spring (each, A228, SF 1.2 on Shigley static limits)

Rate used / by formula	0.125 / 0.117 N/mm
Initial tension used	7.0 N
Body coils, mass	98, 53 g
Safe force (governed by)	40.9 N (hook torsion)
Safe extension (wire) / spec cap	271 / 265 mm
Surge (fixed-fixed)	24 Hz

Launch (chain spring model)

Stroke, ratio, pitch	241 mm, 1.0:1, 35 deg
Spring extension full / rest	265 / 24 mm
Stored energy (2 springs)	12.1 J
Carriage force at full draw	80 N
Muzzle speed	6.85 m/s
Peak acceleration, stroke time	18 g, 59 ms
Efficiency (dart KE / stored)	54 %
Carriage KE at stop, pad peak	1.89 J, 285 N (19 mm pad)
Spring KE at stop, coil clash	0.82 J, 8.0 m/s

Flight (bead-chain tether + reel model, level ground)

Line path at the shot	taut (path only lengthens)
Range at 0.5 / 0.75 / 1.0 N brake	4.18 / 3.76 / 3.39 m (pitch 34)
Same shot, no tether	5.01 m
Flight time, apex (0.75 N)	0.80 s, 1.11 m
Impact speed, angle (0.75 N)	5.7 m/s, 55 deg down
Ballistics table	6 to 47 deg, best 33.5 deg: 3.76 m

Storage reel payout (LIGHT 0.5 N at full reel)

Reel empty mass, inertia full	46 g, 31.4 kg mm2
Line radius full / core	30.7 / 25.0 mm (brake +23 % at core)
Line pull in the stroke (spin-up)	5.05 N
Reel rpm max	2129 rpm
Overrun after landing	0.56 m in 0.27 s

Capstan and tow

Wraps for tow limit at the capstan (57 N, mu 0.15)	2.6 raw, 4 design
Take-up needed for the tow at the actual wraps (7)	0.08 N
Tow at limit: torque, speed, current	3.23 N m, 0.143 m/s, 5.06 A
Wiper at limiter (5 N m), 12.6 V: torque, current, line pull	5.0 N m, 7.0 A, 80 N (break 667 N)
Wiper motor stall (no limiter), 12.6 V: torque, current, line pull	14.1 N m, 17.0 A, 227 N
Tow sensing at the dart: [illegible] per wiper amp above free	N = 14.8 x (I - I_free) - 2.6 N (capstan side x1.15)

Drivetrain (wiper at 11.1 V with sag)

aim: peak torque, current, time	0.33 N m, 1.9 A, 7 s
cock: peak torque, current, time	2.93 N m, 4.7 A, 3 s
reel: peak torque, current, time	0.43 N m, 2.0 A, 107 s
tow: peak torque, current, time	3.23 N m, 5.1 A, 35 s

To measure (then edit base/spec/params.py and run ./run_all.sh)

1. Spring hang test: hang 1, 2, 3 kg from one spring and measure the length each time (hooks included). python calib.py spring FREE_MM 1:MM 2:MM 3:MM, then SPRING k_n_per_mm and F0_n. Every range number scales with the stored energy.
2. Wiper: free speed (turns in 30 s at 12.0 V), free current, stall torque (arm + luggage scale, 1 s) and stall current. python calib.py wiper --v 12 --rpm .. --i-free .. --stall-kg .. --arm-mm .. --i-stall .., then WIPER n0_rpm, stall_nm, stall_a, free_a.
3. Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg), then DART mass_kg.
4. Slip clutch: luggage scale on the line with the reel full: LIGHT 0.5 N, FIRM 5 N (REEL clutch).
5. First shots on level ground: python calib.py range pitch:brake:range ..., then LAUNCH v_scale = FACTOR.